

START A NEW PAGEQUESTION ONE (12 marks)

a) Evaluate

2

$$\int_0^3 \frac{dx}{x^2+9}.$$

b) Find the acute angle between lines $2x - y = 0$ and $x + 3y = 0$.

3

Give your answer correct to the nearest minute.

c) Solve $\frac{x}{2x-1} \leq 5$.

3

d) Differentiate $\cos^4 x$ with respect to x and hence find $\int \sin x \cos^3 x \, dx$.

2

e) A is the point $(-2, -1)$ and B is the point $(1, 5)$.

2

Find the coordinates of Q which divides AB internally in the ratio 5:3.START A NEW PAGEQUESTION TWO (12 marks)a) Evaluate $\int_0^{\frac{\pi}{3}} \sin^2(2x) \, dx$.

3

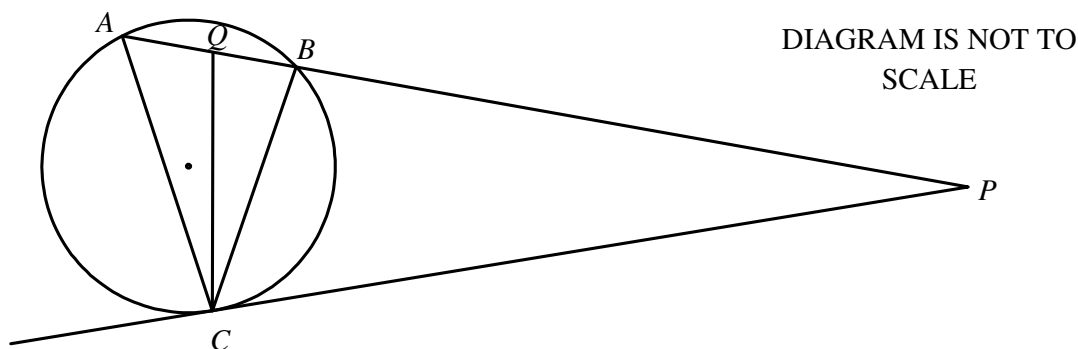
b) Use the substitution $u = \tan x$, to evaluate $\int_0^{\frac{\pi}{3}} \tan^2 x \sec^2 x \, dx$.

3

- c) A biased coin has a probability of 0.6 of showing a head when it is tossed.
The biased coin is tossed five times.
- (i) What is the probability of tossing exactly three heads and two tails? 2
- (ii) What is the probability of getting less than three heads? 2
- (iii) What is the most likely outcome and what is the probability that this event will occur? 2

START A NEW PAGEQUESTION THREE (12 marks)

- a) Find the exact value of $\sin\left(\sin^{-1}\left(\frac{4}{5}\right) - \tan^{-1}\left(\frac{-5}{12}\right)\right)$. 3
- b) In the diagram below PC is a tangent to the circle at C and QC bisects $\angle ACB$.



- (i) Copy the diagram into your answer booklet and prove that $PC = PQ$. 2
- (ii) If $PC = 10\text{cm}$ and $AQ = 2\text{cm}$, Find the length of QB . 2

- c) Molten plastic at a temperature of 250°C , is poured into a mould to form a car part. After 20 minutes the plastic has cooled to 150°C . If the temperature after t minutes, is $T^{\circ}\text{C}$, and the surrounding temperature is 30°C , then the rate of cooling is given by:

$$\frac{dT}{dt} = -k(T - 30), \text{ where } k \text{ is constant.}$$

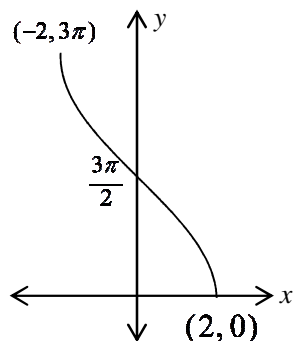
- (i) Show that $T = 30 + Ae^{-kt}$, where A is a constant, satisfies this equation. 1
- (ii) Show that the value of A is 220°C . 1
- (iii) Find the value of k to 2 decimal places. 1
- (iv) The plastic can be taken out of the mould when the temperature drops below 80°C . How long after the plastic has been poured will this temperature be reached? Give your answer to the nearest minute. 2

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QUESTION FOUR (12 marks)

- a) Prove by mathematical induction that:
 $5^n - 4n - 1 \geq 0$, for all $n \geq 1$. 4
- b) John contributes \$800 every quarter into a superannuation fund earning 6% pa, compounded quarterly. Contributions are made for 10 years and then John stops making contributions for 3 years whilst unemployed. He is re-employed, and contributions resume under the previous conditions. John works for a further 7 years until retirement.
- (i) Show that on retirement the contributions for the first 10 years can be expressed as:
 $800(1.015)^{80} + 800(1.015)^{79} + \dots + 800(1.015)^{41}$. 1
- (ii) Hence, calculate the total amount in the superannuation fund on retirement. 3

- c) Below is the graph $y = a \cos^{-1} bx$.



- (i) Show that the value of a is 3 and that b is $\frac{1}{2}$. 2
- (ii) Find the exact value of the area enclosed by the curve $y = a \cos^{-1} bx$, the x axis and the line $y = \frac{\pi}{2}$. 2

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QUESTION FIVE (12 marks)

- a) A particle moves in a straight line so that its acceleration, a , is given by $a = 4x$. The displacement, x , of the particle is initially 1 metre to the right of the origin with a velocity of -2ms^{-1} .
- (i) Show that $v = -2x \text{ ms}^{-1}$. 2
- (ii) Express x as a function of t . 2
- (iii) Hence, find the displacement when $t = 2$ seconds to 3 decimal places. 1

- b) (i) Write an expression for $(1+x)^{20} + (1-x)^{20}$. 2
Leave your answer in nC_k notation.

(ii) Show that $\sum_{k=0}^{10} {}^{20}C_{2k} x^{2k} = \frac{1}{2}[(1+x)^{20} + (1-x)^{20}]$. 1

- (iii) Integrate both sides of the equation in part (ii). 2

- (iv) Hence, or otherwise show:

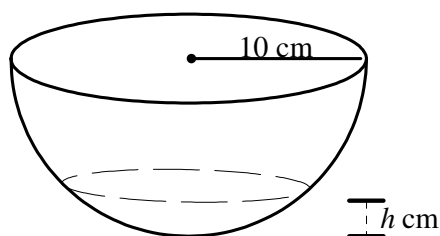
$${}^{20}C_0 + \frac{1}{3} {}^{20}C_2 + \frac{1}{5} {}^{20}C_4 + \dots + \frac{1}{21} {}^{20}C_{20} = \frac{2^{20}}{21}. \quad 2$$

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QUESTION SIX (12 marks)

- a) (i) The diagram below represents a hemispherical bowl of radius 10 cm. It is filled with water to a depth of h cm. By finding the volume generated by rotating $x^2 + y^2 = 100$ between $y=10$ and $y=10-h$ about the y axis, show that the volume of water in the bowl is given by:

$$V = \frac{\pi h^2}{3}(30-h) \quad 2$$



- (ii) The hemispherical bowl referred to above is being filled with water at a constant rate of $2\pi \text{ cm}^3 \text{ min}^{-1}$. Find the rate of increase of the depth of the water when the depth is 2 cm. 3

b) Consider the function $f(x) = \sqrt{x+3}$

(i) Find an expression for $y = f^{-1}(x)$. 1

(ii) State the domain for $y = f^{-1}(x)$. 1

(iii) $f(x)$ and $f^{-1}(x)$ intersect at exactly one point P . Let α be the x coordinate of P . 2
Explain why α is a root of the equation $x - \sqrt{x+3} = 0$. 1

(iv) Show that the value of α occurs when $2 < \alpha < 3$. 1

(v) Take $x = 2.5$ as the first approximation of α , use one application of Newton's 2
method to find a second approximation for α correct to 3 decimal places.

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QUESTION SEVEN (12 marks)

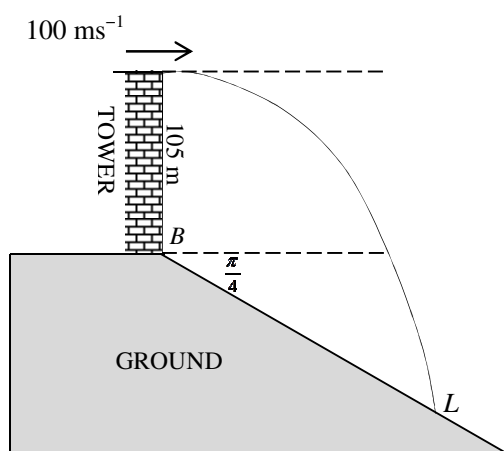
a) On a certain day the depth of water in a harbour is 7 metres at low tide (9:25am) and $10\frac{2}{3}$ metres at high tide (3:40pm). Assume the rise and fall of the surface of the water to be in simple harmonic motion in the form $\ddot{x} = -n^2(x - b)$ where $x = b$ is the centre of motion and $x = a$ is the amplitude.

(i) Show that $x = b - a \cos nt$ satisfies $\ddot{x} = -n^2(x - b)$. 1

(ii) Find the values of a , b and n . 3

(iii) Hence, find the earliest time before 3:40pm on this day, a boat may safely enter the 2
harbour if the minimum depth of $9\frac{1}{2}$ metres of water is required.

- b) A bullet is fired horizontally with a velocity of 100ms^{-1} from the top of a tower 105 metres high. The tower is at the top of a hill, which slopes downwards at an angle of depression of $\frac{\pi}{4}$. The bullet lands at L , on the ground as represented on the diagram below.



- (i) Consider B , the base of the tower, as the origin, and using acceleration due to gravity as 10 ms^{-2} , show that the expressions for the x and y coordinates of the position of the bullet at time t seconds are: 2
- $$x = 100t \text{ and } y = 105 - 5t^2.$$
- (ii) Show that the equation of the line BL is $y = -x$. 1
- (iii) Find the time taken for the bullet to hit the ground at L . 2
- (iv) Find the distance BL to the nearest metre. 1

END OF THE EXAMINATION

QUESTION ONE (12 marks)

a) Evaluate

$$\int_0^3 \frac{dx}{x^2+9} = \left[\frac{1}{3} \tan^{-1} \frac{x}{3} \right]_0^3 = \frac{\pi}{12}$$

b)

$$m_1 = 2 \text{ and } m_2 = -\frac{1}{3}$$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| = 7$$

$$\theta = 81^\circ 52'$$

c)

$$\frac{x}{2x-1} \times (2x-1)^2 \leq 5(2x-1)^2 \quad x \neq \frac{1}{2}$$

$$2x^2 - x \leq 20x^2 - 20x + 5$$

$$0 \leq 18x^2 - 19x + 5$$

$$0 \leq (9x-5)(2x-1)$$

$$x \geq \frac{5}{9} \text{ and } x < \frac{1}{2}$$

d)

$$\frac{dy}{dx} = 4 \cos^3 x \times -\sin x$$

$$\frac{dy}{dx} = -4 \cos^3 x \sin x$$

$$-4 \int \cos^3 x \sin x dx = \cos^4 x + c$$

$$\int \cos^3 x \sin x dx = -\frac{1}{4} \cos^4 x + c$$

e)

$$\left(\frac{-6+5}{8}, \frac{-3+25}{8} \right)$$

$$\left(\frac{-1}{8}, 2\frac{3}{4} \right)$$

QUESTION TWO (12 marks)

a)

$$\begin{aligned}
 \int_0^{\frac{\pi}{3}} \sin^2(2x) dx &= \int_0^{\frac{\pi}{3}} \frac{1 - \cos 2(2x)}{2} dx \\
 &= \frac{1}{2} \int_0^{\frac{\pi}{3}} 1 - \cos(4x) dx = \frac{1}{2} \left[x - \frac{1}{4} \sin(4x) \right]_0^{\frac{\pi}{3}} \\
 &= \frac{\pi}{6} + \frac{\sqrt{3}}{16}
 \end{aligned}$$

b)

$$u = \tan x$$

$$\frac{du}{dx} = \sec^2 x \quad du = \sec^2 x \, dx \quad \text{when } x = 0 \quad u = \tan 0 = 0$$

$$\text{when } x = \frac{\pi}{3} \quad u = \tan \frac{\pi}{3} = \sqrt{3}$$

$$\int_0^{\sqrt{3}} u^2 du = \left[\frac{u^3}{3} \right]_0^{\sqrt{3}} = \frac{3\sqrt{3}}{3} = \sqrt{3}$$

c)

$$(i) \quad {}^5_2C(0.4)^2(0.6)^3 = 0.3456$$

$$(ii) \quad {}^5_3C(0.4)^3(0.6)^2 + {}^5_4C(0.4)^4(0.6)^1 + {}^5_5C(0.4)^5(0.6)^0 = 0.31744$$

(iii) The most likely outcome is 3 heads and 2 tails which has 0.3456 chance of occurring.

QUESTION THREE (12 marks)

a) $\sin\left(\sin^{-1}\left(\frac{4}{5}\right) - \tan^{-1}\left(\frac{-5}{12}\right)\right).$

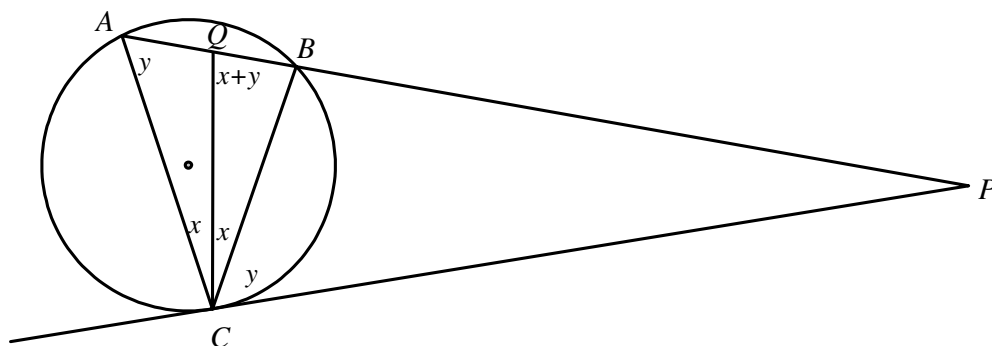
$$\sin(A - B) = \sin A \cos B - \sin B \cos A$$

$$= \left(\frac{4}{5} \times \frac{-12}{13}\right) - \left(\frac{5}{13} \times \frac{3}{5}\right)$$

$$= \frac{-63}{65}$$

b)

(i)



Let $\angle ACQ = x$, $\angle QCB = x$ given that QC bisect $\angle ACB$

Let $\angle BCP = y$, $\angle CAB = y$ (angle between chord and tangent equal angle in the alternate segment)

$\angle CQP = x + y$ (external angle in $\triangle AQC$ is equal to the sum of the opposite internal angles)

$\angle QCP = x + y$

$\triangle PQC$ is an isosceles $\triangle$. ($\angle CQP = \angle QCP$)

$QP = PC$ (sides opposite equal angles are equal)

(ii)

$$PC^2 = PB \times PA$$

$$\text{Let } PB = x, QB = 10 - x$$

$$10^2 = x \times 12$$

$$x = 100 \div 12 = 8\frac{1}{3}$$

$$QB = 10 - 8\frac{1}{3} = 1\frac{2}{3}$$

c)

(i)

$$T = 30 + Ae^{-kt}$$

$$Ae^{-kt} = T - 30$$

$$\frac{dT}{dt} = -kAe^{-kt} = -k(T - 30)$$

(ii) Show that the value of A is 220°C .

$$T_0 = 250$$

$$250 = 30 + A$$

$$A = 220$$

(iii) Find the value of k to 2 decimal places.

$$150 = 30 + 220e^{-20k}$$

$$120 = 220e^{-20k}$$

$$\frac{120}{220} = e^{-20k}$$

$$k = \frac{1}{20} \ln \frac{11}{6} = 0.03$$

(iv)

$$80 = 30 + 220e^{-0.03t}$$

$$50 = 220e^{-0.03t}$$

$$\frac{50}{220} = e^{-0.03t}$$

$$t = \frac{\ln \frac{5}{22}}{-0.03} = 49 \text{ minutes}$$

QUESTION FOUR (12 marks)

- a) Prove by mathematical induction that:

$$5^n - 4n - 1 \geq 0, \text{ for all } n \geq 1.$$

Test for $n = 1$

$$5^1 - 4(1) - 1 \geq 0,$$

$$0 \geq 0$$

True for $n = 1$.Assume true for $n = k$, where $k \geq 1$

$$\text{assume } 5^k - 4k - 1 \geq 0,$$

Test for $n = k + 1$

$$5^{k+1} - 4(k+1) - 1$$

$$= 5 \times 5^k - 4k - 4 - 1$$

$$= 5 \times 5^k - 20k + 16k - 5$$

$$= 5(5^k - 4k - 1) + 16k$$

Since $(5^k - 4k - 1) \geq 0$ and $16k \geq 0$ then $5(5^k - 4k - 1) + 16k \geq 0$, where $k \geq 1$ Since true for $n = 1$, and true for $n = k + 1$,then by the process of mathematical induction, it is true for all values of n .

- b)

- (i)

There are 10 years of contributions, then 3 years without contributions, followed by a further 7 years of contribution. The first contribution occurred 20 years before the last.

Contributions occurred quarterly (80 quarters @ 1.5% per quarter)

$$\text{First contribution} = 800(1.015)^{80}$$

$$\text{Second contribution} = 800(1.015)^{79}$$

$$\text{Third contribution} = 800(1.015)^{78}$$

$$\text{Last contribution before the 3 year break} = 800(1.015)^{41}$$

Therefore total contribution made for first 10 years is

$$800(1.015)^{80} + 800(1.015)^{79} + \dots + 800(1.015)^{41}.$$

- (ii) Hence, calculate the total amount in the superannuation fund on retirement.

Total contribution = first 10 years contribution + Last 7 years contribution

$$= 800(1.015)^{80} + 800(1.015)^{79} + \dots + 800(1.015)^{41} + 800(1.015)^{28} + 800(1.015)^{27} + \dots + 800(1.015)^1.$$

$$\frac{800(1.015)(1.015^{28} - 1)}{1.015 - 1} + \frac{800(1.015)^{41}(1.015^{40} - 1)}{1.015 - 1}$$

$$\$27998.96 + \$79935.68$$

$$= \$107934.64$$

c)

- (i) Show that the value of a is 3 and that b is $\frac{1}{2}$.

$$y = a \cos^{-1} bx$$

$$x = a \cos^{-1} by$$

$$\frac{x}{a} = \cos^{-1} by$$

$$\cos \frac{x}{a} = by$$

$$\frac{1}{b} \cos \frac{x}{a} = y$$

$$\text{amplitude} = 2. \text{ Therefore } \frac{1}{b} = 2, b = \frac{1}{2}$$

$$\text{period} = 6\pi. \text{ Therefore } \frac{1}{a} = \frac{2\pi}{6\pi}, a = 3$$

(ii)

$$\int_0^{\frac{\pi}{2}} 2 \cos \frac{x}{3} dx = 2 \left[3 \sin \frac{x}{3} \right]_0^{\frac{\pi}{2}}$$

$$6 \sin \frac{\pi}{6} = 3 \text{ units}^2$$

QUESTION FIVE (12 marks)

a) (i)

$$a = \ddot{x} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$$

$$\frac{1}{2} v^2 = \int 4x dx$$

$$\frac{1}{2} v^2 = 2x^2 + c$$

when $x = 1$, $v = -2$, therefore $c = 0$

$$v^2 = 4x^2$$

$v = \pm \sqrt{4x^2}$ Since velocity is initially negative then

$$v = -\sqrt{4x^2} = -2x$$

(ii) Express x as a function of t .

$$v = \frac{dx}{dt} = -2x$$

$$\frac{dt}{dx} = \frac{1}{-2x}$$

$$t = \int \frac{1}{-2x} dx = -\frac{1}{2} \ln x + c$$

when $t = 0$, $x = 1$, $\therefore c = 0$

$$t = -\frac{1}{2} \ln x$$

$$-2t = \ln x$$

$$x = e^{-2t}$$

(iii) Hence, find the displacement when $t = 2$ seconds to 3 decimal places.

when $t = 2$

$$x = e^{-4} = 0.018$$

b) (i)

$$(1+x)^{20} + (1-x)^{20}$$

$$= {}^{20}_0 C + {}^{20}_1 Cx + {}^{20}_2 Cx^2 + \dots + {}^{20}_{20} Cx^{20} + {}^{20}_0 C - {}^{20}_1 Cx + {}^{20}_2 Cx^2 - \dots + {}^{20}_{20} Cx^{20}$$

$$= 2 \left({}^{20}_0 C + {}^{20}_2 Cx^2 + {}^{20}_4 Cx^4 + \dots + {}^{20}_{20} Cx^{20} \right)$$

$$= 2 \sum_{k=0}^{10} {}^{20}_{2k} Cx^{2k}$$

(ii)

$$(1+x)^{20} + (1-x)^{20} = 2 \sum_{k=0}^{10} {}^{20}_{2k} C x^{2k}$$

$$\sum_{k=0}^{10} {}^{20}_{2k} C x^{2k} = \frac{1}{2} \left((1+x)^{20} + (1-x)^{20} \right)$$

(iii) Integrate both sides of the equation in part (ii).

$$\frac{1}{2} \left((1+x)^{20} + (1-x)^{20} \right) = \sum_{k=0}^{10} {}^{20}_{2k} C x^{2k}$$

$$\frac{1}{2} \frac{1}{21} \left((1+x)^{21} - (1-x)^{21} \right) + c = \sum_{k=0}^{10} \frac{1}{2k+1} {}^{20}_{2k} C x^{2k+1} + c$$

(iv) Hence, or otherwise show:

$${}^{20}_0 C_0 + \frac{1}{3} {}^{20}_2 C_2 + \frac{1}{5} {}^{20}_4 C_4 + \dots + \frac{1}{21} {}^{20}_{20} C_{20} = \frac{2^{20}}{21}.$$

Let $x = 10 \therefore c = 0$ When $x = 1$

$$\frac{1}{2} \frac{1}{21} \left((1+1)^{21} - (1-1)^{21} \right) = \sum_{k=0}^{10} \frac{1}{2k+1} {}^{20}_{2k} C 1^{2k+1}$$

$$\frac{1}{21} \frac{1}{2} \left(2^{21} \right) = \frac{1}{21} \left(2^{20} \right) = \sum_{k=0}^{10} \frac{1}{2k+1} {}^{20}_{2k} C$$

$$\frac{2^{20}}{21} = \sum_{k=0}^{10} \frac{1}{2k+1} {}^{20}_{2k} C = \left({}^{20}_0 C + \frac{1}{3} {}^{20}_2 C x^2 + \frac{1}{5} {}^{20}_4 C x^4 + \dots + \frac{1}{21} {}^{20}_{20} C x^{20} \right)$$

QUESTION SIX (12 marks)

(i)

$$x^2 + y^2 = 100$$

$$x^2 = 100 - y^2$$

$$V = \pi \int_{10-h}^{10} 100 - y^2 dy$$

$$V = \pi \left[100y - \frac{1}{3}y^3 \right]_{10-h}^{10}$$

$$V = \pi \left[\left(100(10) - \frac{1}{3}(10)^3 \right) - \left(100(10-h) - \frac{1}{3}(10-h)^3 \right) \right]$$

$$V = \pi \left[\left(1000 - \frac{1}{3}(1000) \right) - \left(1000 - 100h - \frac{1}{3}(1000 - 300h + 30h^2 - h^3) \right) \right]$$

$$V = \frac{\pi}{3} [30h^2 - h^3]$$

$$V = \frac{\pi h^2}{3} [30 - h]$$

(ii)

$$\frac{dV}{dt} = 2\pi$$

$$V = \frac{\pi h^2}{3} [30 - h] = 10\pi h^2 - \frac{\pi h^3}{3}$$

$$\frac{dV}{dh} = 20\pi h - \pi h^2$$

$$\frac{dh}{dt} = \frac{dh}{dV} \times \frac{dV}{dt}$$

$$\frac{dh}{dt} = \frac{1}{20\pi h - \pi h^2} \times 2\pi$$

$$\text{when } h = 2$$

$$\frac{dh}{dt} = \frac{1}{40\pi - 4\pi} \times 2\pi = \frac{2\pi}{36\pi} = \frac{1}{18} \text{ cm min}^{-1}$$

b)

- (i) Find an expression for
- $y = f^{-1}(x)$
- .

$$x = \sqrt{y+3}$$

$$x^2 = y+3$$

$$y = x^2 - 3$$

$$f^{-1}(x) = x^2 - 3$$

(ii)

Domain $f^{-1}(x)$ is the range for $f(x)$.

$$\text{For } f(x) = \sqrt{x+3} \quad x \geq 0$$

(iii)

$$f^{-1}(f(x)) = f(f^{-1}(x)) = x$$

$$y = x \quad \text{and} \quad y = \sqrt{x+3}$$

$$y = y$$

$$x = \sqrt{x+3}$$

$$x - \sqrt{x+3} = 0$$

- (iv) Show that the value of
- α
- occurs when
- $2 < \alpha < 3$
- .

$$\text{when } x = 2 \quad 2 - \sqrt{2+3} < 0$$

$$\text{when } x = 3 \quad 3 - \sqrt{3+3} > 0$$

 $\therefore$ there will be a zero between 2 and 3.

- (v) Take
- $x = 2.5$
- as the first approximation of
- α
- , use one application of Newton's

method to find a second approximation for α correct to 3 decimal places.

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$x_1 = 2.5 - \frac{f(2.5)}{f'(2.5)}$$

$$x_1 = 2.303$$

QUESTION SEVEN (12 marks)

a)

- (i) Show that
- $x = b - a \cos nt$
- satisfies
- $\ddot{x} = -n^2(x - b)$
- .

$$x = b - a \cos nt \quad \therefore a \cos nt = b - x$$

$$\frac{dx}{dt} = -an \sin nt$$

$$\frac{d^2x}{dt^2} = an^2 \cos nt = n^2(a \cos nt) = n^2(b - x) = -n^2(x - b)$$

$$\ddot{x} = \frac{d^2x}{dt^2} = -n^2(x - b)$$

- (ii) Find the values of
- a
- ,
- b
- and
- n
- .

$$b = 8\frac{5}{6} \quad a = 1\frac{5}{6}$$

$$\frac{2\pi}{n} = 12\frac{1}{2} \text{ hours}$$

$$n = \frac{4\pi}{25}$$

- (iii) Hence, find the earliest time before 3:40pm on this day, a boat may safely enter the

harbour if the minimum depth of $9\frac{1}{2}$ metres of water is required.

$$x = 8\frac{5}{6} - 1\frac{5}{6} \cos \frac{4\pi t}{25}$$

$$\text{when } x = 9\frac{1}{2}$$

$$9\frac{1}{2} = 8\frac{5}{6} - 1\frac{5}{6} \cos \frac{4\pi t}{25}$$

$$\cos \frac{4\pi t}{25} = \frac{-4}{11}$$

$$\frac{4\pi t}{25} = \cos^{-1}\left(\frac{-4}{11}\right)$$

$$t = 3 \text{ hours } 51 \text{ minutes } 56 \text{ seconds}$$

$$9:25 \text{ am} + 3 \text{ hours } 51 \text{ minutes } 56 \text{ seconds} = 1:17 \text{ pm}$$

b)

(i)

$$\begin{aligned}\ddot{x} &= 0 & \ddot{y} &= -g \\ \dot{x} &= 100 & \dot{y} &= -gt + c & c &= 0 \\ x &= 100t & y &= \frac{-gt^2}{2} + c & \text{when } t=0 & y=105 \therefore c=105 \\ & & y &= \frac{-gt^2}{2} + 105 = \frac{-10t^2}{2} + 105 = -5t^2 + 105\end{aligned}$$

(ii) Show that the equation of the line BL is $y = -x$.

$$m = \tan \theta$$

gradient of BL is -1

$$\text{When } x=0, y=0$$

$$y = -x$$

(iii) Find the time taken for the bullet to hit the ground at L .

$$x = 100t \quad y = -5t^2 + 105$$

$$y = -5\left(\frac{x}{100}\right)^2 + 105$$

$$y = \left(\frac{-5x^2}{10000}\right) + 105$$

$$y = -x$$

$$-x = \frac{-5x^2}{10000} + 105$$

$$x^2 - 2000x - 210000 = 0$$

$$(x - 2100)(x + 100) = 0$$

$$x = 2100$$

$$t = \frac{2100}{100} = 21 \text{ minutes}$$

(iv) Find the distance BL to the nearest metre.

$$BL = \sqrt{2100^2 + 2100^2}$$

$$BL = 2970 \text{ cm}$$

$$BL \approx 30 \text{ m}$$